

MIRROR

Multimodal Intelligent Radiology Reasoning and Observation Reporter

A grounded, explainable chest-radiograph pipeline that predicts fourteen thoracic pathologies, localizes the visual evidence behind each finding, and drafts a clinician-style report, with predictive quality and explanation quality measured side by side.

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Abstract. Most medical-imaging models stop at a prediction, offering neither the location of the evidence nor an account a clinician can read. We present **MIRROR**, a three-layer chest-radiograph pipeline that composes (i) a multi-label disease classifier, (ii) class-activation-based evidence localization, and (iii) an evidence-grounded natural-language report generator into a single traceable system. The defining design choice is that each layer’s output is the grounded input to the next, and the language layer never sees pixels, so every sentence in the generated report traces back to a predicted probability and a saliency region, and the reasoning layer cannot invent findings the classifier did not produce. We pair the system with an evaluation framework that measures predictive quality (AUROC/F1) and explanation quality (pointing game, localization IoU) side by side, and an ablation that isolates the cost of the added interpretability layers. Because those layers are strictly post-hoc, predictive metrics are provably invariant to them, which lets us frame the contribution as *interpretability at no predictive cost*. All predictive numbers carry bootstrap 95% confidence intervals and can be aggregated across training seeds, and every run stamps a reproducibility record. **[PENDING: fold in a one-sentence quantitative headline (macro AUROC, localization accuracy, and the ablation’s invariance result) once the classifier is trained on ChestX-ray14.]**

1 Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet adoption in the clinic remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it reached its conclusion. High-stakes decisions demand models that can be interrogated, not merely queried [15]. Post-hoc saliency methods such as Grad-CAM [9] and Score-CAM [10] expose *where* a prediction comes from, and large language models can render structured evidence as prose, but these capabilities are rarely composed into one system whose outputs remain mutually consistent and auditable end to end.

MIRROR integrates all three into a single pipeline:

Image → Prediction → Localization → Reasoning → Report.

Its central design constraint is *grounding*: the language layer is given only the structured evidence produced upstream (labels, probabilities, and named saliency regions), never the raw image. This makes hallucinated findings structurally impossible, since the report can only discuss what the classifier predicted and the localizer situated, and it makes the whole chain traceable.

Research question. Can a multimodal system that combines image classification, visual explainability, and language

generation improve interpretability without degrading predictive performance, relative to a classification-only baseline? We build the system so this question is *measurable*: predictive and explanation metrics live side by side, and an ablation reproduces the classification-only baseline the question names.

Contributions.

1. A three-layer, evidence-grounded radiology pipeline in which the language layer reasons only over structured upstream evidence, so every report sentence is traceable to a probability and a saliency region.
2. A grounding constraint that makes the generated report auditable and non-hallucinatory *by construction*, together with a post-hoc layer design whose invariance to the predictions is verified empirically.
3. An open-source implementation spanning interchangeable backbones (DenseNet-121, EfficientNet-B0, ViT-B/16), two explainability methods (Grad-CAM, Score-CAM), an LLM report backend with a deterministic offline fallback, native DICOM ingest, and a web front end served by two interchangeable inference engines behind one response contract (Fig. 2).
4. An evaluation framework that scores prediction and explanation quality jointly, with bootstrap confidence intervals,

multi-seed aggregation, and an ablation isolating the predictive cost of interpretability, so the research question can be answered quantitatively. **[PENDING: fold in the measured result once the model is trained.]**

2 Literature Review

Datasets and classification. Large, labeled chest-radiograph corpora made supervised deep learning on this modality practical. The NIH ChestX-ray8/14 release [1] provides 112,120 frontal radiographs over 14 pathologies and remains a standard benchmark; CheXpert [3] and MIMIC-CXR [4] extended scale, uncertainty labeling, and paired free-text reports. On these benchmarks CheXNet [2] showed a 121-layer DenseNet [5] reaching radiologist-level pneumonia detection. Convolutional backbones such as EfficientNet [6] and, more recently, Vision Transformers [7] now form the standard toolkit; MIRROR keeps all three interchangeable and adopts DenseNet-121 as its primary configuration for comparability.

Visual explanation. Class Activation Mapping [8] localized the evidence behind a CNN prediction; Grad-CAM [9] generalized it to arbitrary architectures via gradients, and Score-CAM [10] removed the gradient dependence with a perturbation-based weighting. Model-agnostic attribution methods, notably LIME [11] and SHAP [12], explain individual predictions without access to model internals. To move from qualitative heatmaps to measurable evidence, the pointing-game protocol [13] and intersection-over-union against annotated lesions provide quantitative localization scores, which MIRROR reports directly.

Rigor and critique of saliency. Explanations are only useful if they are faithful. Adebayo et al. [14] showed several saliency methods can be insensitive to model and data, and argued for sanity checks; Rudin [15] cautioned against post-hoc explanation of black boxes for high-stakes decisions; and Doshi-Velez and Kim [16] called for a rigorous, measurable science of interpretability. These critiques motivate two of MIRROR’s design choices: quantitative localization evaluation against ground-truth boxes, and hard grounding so the language layer cannot outrun the evidence.

Report generation. Systems that emit radiology prose range from TieNet [17], which jointly embeds image and text, to memory-driven transformers such as R2Gen [18], to clinically focused decoders optimized for factual correctness [19]. A recurring failure mode is hallucination: prose generated directly from pixels can assert findings the underlying model never detected. MIRROR sidesteps this by restricting generation to structured, verified evidence, trading open-ended fluency for auditability.

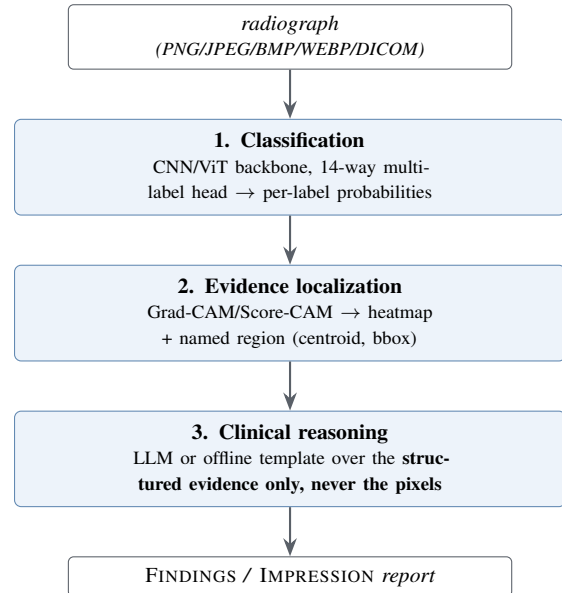


Figure 1: The MIRROR pipeline. Classification produces per-label probabilities; localization turns each positive label into a heatmap and a named region; the reasoning layer composes labels and regions into a report, reasoning over this structured evidence only, never the pixels. Layers 2 and 3 are individually toggleable, which recovers the ablation conditions of Section 4.

Explainable AI in medicine and positioning. Surveys of explainable AI for medical imaging [20] note that interpretability and report generation are usually studied in isolation. MIRROR’s contribution is not a new classifier or a new saliency method, but their *grounded composition* into one traceable pipeline, together with a framework that measures whether adding the explanation and reasoning layers helps interpretability without a predictive penalty.

3 System Architecture

MIRROR chains three layers; each layer’s output is the grounded input to the next (Fig. 1).

3.1 Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a shared 14-way multi-label head (sigmoid applied at loss and inference time, so the network emits raw logits). Inputs are resized to 224×224 and normalized with ImageNet statistics. Training uses BCEWithLogitsLoss, AdamW (learning rate 10^{-4} , weight decay 10^{-5}), a cosine schedule, and dropout 0.2; checkpoints are selected on best validation macro-AUROC. The 14 labels follow the canonical NIH ChestX-ray14 taxonomy, kept as a single source of truth shared by the classifier head, the report generator, and the evaluation scripts. Because real radiology data arrives as DICOM rather than PNG/JPEG, a dedicated ingest applies

the modality LUT (rescale slope/intercept), the VOI (window) LUT, and MONOCHROME1 inversion before handing the pipeline a display-ready image, and lifts only non-PHI technical tags (modality, view position, laterality).

3.2 Layer 2: Evidence Localization

For each positive label (top- k , $k=3$ by default, to bound compute), the explainability layer computes a class-activation map, either Grad-CAM (forward/backward hooks on a resolved target layer) or Score-CAM (gradient-free, perturbation-based), and reduces it to a named region so the evidence can be referred to in words. Target layers are resolved per backbone; ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 spatial grid. The activated region’s centroid is mapped into a 3×3 anatomical grid in radiology convention (patient-right on the viewer’s left), yielding a plain-English location such as “the right lower zone,” and a colormap overlay is rendered for the UI (Fig. 3, left).

3.3 Layer 3: Clinical Reasoning

The reasoning layer converts structured evidence (labels, probabilities, and named regions) into a FINDINGS/IMPRESSION report via a system prompt plus an evidence-grounded user prompt. Per-label plain-English descriptions ground the model so it need not recall what each token means. The backend is an LLM (Claude) with a deterministic offline template fallback so the system always yields a coherent report even with no network or API key; generation runs at low temperature for stability, and the LLM backend falls back to the template on any error. Predictions below the confidence threshold (0.5) are surfaced as pertinent negatives rather than silently dropped. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, never *what* it may say; and because the layer never receives the image, it cannot introduce findings absent from the upstream evidence (Fig. 3, right).

3.4 Orchestration and Serving

A single `analyze()` entry point runs the stages and returns one result object (predictions, overlays, report) consumed by both the REST API and the notebooks; the two later layers are individually toggleable, which recovers the ablation conditions of Section 4 and records per-stage wall-clock timings. Two interchangeable serving engines satisfy the *same* JSON response contract (Table 1), so the web UI (Fig. 2) is identical regardless of which engine answered: a finding carries *either* a rendered Grad-CAM overlay (local) *or* a normalized bounding box (hosted), and the same viewer draws whichever is present. The hosted engine exists because the PyTorch pipeline cannot fit typical serverless limits (no GPU, a small bundle ceiling, cold-start model loads); it substitutes a vision LLM constrained to a structured tool schema that emits

Table 1: Two interchangeable inference engines behind one response contract.

	Local stack	Hosted (serverless)
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores 14 labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM (same prompt shape)
Inputs	+DICOM, BMP	PNG/JPEG/WEBP
Needs	Python, ~6 GB	one API-key env var

the identical JSON, and with no API key returns a clearly-labelled deterministic demo result so the deployed site never hard-fails.

4 Experimental Setup

The evaluation is designed to answer the research question’s two halves, prediction and explanation, with parallel harnesses that emit machine-readable summaries the tables below are generated from. *The design is fixed; the numbers await a trained checkpoint.*

Data. We use NIH ChestX-ray14 [1]: 112,120 frontal radiographs from 30,805 patients, labeled across 14 pathologies (multi-label), with the official train/validation and held-out test splits.

Localization ground truth. We score heatmaps against the official `BBox_List_2017.csv`: ~984 hand-drawn boxes over 8 of the 14 pathologies (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax). Boxes are given in original 1024×1024 pixel coordinates and rescaled into the model’s heatmap grid; the remaining six labels have no box ground truth and are not scored.

Predictive metrics. Per-label and macro AUROC, plus macro F1 at a 0.5 threshold, on the official test split. Each metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement $B=1000$ times, and a *single* resample drives all metrics at once, so the intervals are mutually consistent. For a statistic θ with bootstrap replicates $\{\hat{\theta}_b^*\}$ we report the two-sided percentile interval $[\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^*]$ with $\alpha=0.05$, which supports statements of the form “config A beats config B.”

Explanation metrics. Following [13] and the ChestX-ray8 localization protocol [1]: (i) *pointing game*, does the heatmap’s peak fall inside the lesion box; (ii) *mean IoU*, overlap between the box and the heatmap thresholded at 0.5; and (iii) *localization accuracy*, the fraction of boxes with $\text{IoU} \geq 0.1$ (the T(IoU) measure).

*MIRROR reading-room UI screenshot
(add image to paper/figures/)*

Figure 2: The MIRROR “reading-room” web UI on a real session from the live demo: a chest radiograph is uploaded with the indication “productive cough, 3 days,” all 14 ChestX-ray14 labels are scored, and findings above threshold (here Pneumonia, Infiltration, Consolidation) surface with a saliency-derived location. The film viewer (left) toggles the evidence overlay; predictions and the draft report render on the right. Both serving engines produce this identical interface.

Ablation. Three conditions isolate the effect of the added layers (Table 4): `classification_only`, `with_localization`, and `full_mirror`. Because layers 2–3 are post-hoc, the predictive metrics are expected to be identical across conditions; the harness verifies this empirically and reports the maximum prediction delta, so any change in interpretability can be attributed cleanly to added capability rather than a different model.

Reproducibility. Runs are seeded (default 42); robustness to training stochasticity is summarized across seeds {0, 1, 2} as mean $\pm$ std, reported alongside the per-model bootstrap CIs. Every results file stamps a reproducibility record (seed, git commit, library versions).

5 Results

[PENDING: Every number in Tables 2–4 and in Figures 4–5 is shown in red because it is an illustrative, literature-scale placeholder taken from the committed results/evaluation/.json snapshots (each stamped with an _note field stating exactly this); the values document the OUTPUT FORMAT of each harness, not a measured benchmark. Once models/checkpoints/densenet121.best.pt exists, run evaluation/{evaluate, evaluate.localization, ablation, aggregate_seeds}.py, each table and graph regenerates from the resulting JSON. Replace the red values with the measured ones and set them black.]*

The qualitative behavior is already visible end to end: Fig. 3 shows, for one real study, the Grad-CAM evidence overlay on the film and the grounded FINDINGS/IMPRESSION draft it produces. The quantitative tables and graphs below report the three harnesses’ outputs in their released format. Figure 4 visualizes the predictive half (per-label AUROC with bootstrap CIs) and Figure 5 the explanation half (per-lesion localization), the two sides of the research question.

The ablation makes the headline argument explicit: the macro AUROC/F1 columns are constant across all three rows (maximum probability delta = 0), so interpretability is added at no predictive cost, while the latency column shows the

cost is instead a modest, bounded increase in wall-clock time (dominated by the localization pass).

6 Discussion

[PENDING: Interpret the tables and graphs once real numbers are available. Anticipated storyline: (1) the classifier reaches the CheXNet-range AUROC expected on ChestX-ray14; (2) the localization results substantiate the novelty claim that the heatmaps land on annotated lesions; (3) the ablation shows the added layers leave AUROC/F1 unchanged; (4) contrast Grad-CAM vs. Score-CAM localization.]

Independent of the trained numbers, MIRROR’s grounding constraint is a qualitative contribution. Because the language layer sees only structured evidence, the generated report is auditable by construction and cannot assert findings the classifier did not produce; this trades open-ended fluency for a form of safety that matters in a clinical setting. The post-hoc placement of layers 2–3 is likewise not merely convenient: because the classifier’s logits are untouched, any change in a predictive metric across ablation conditions would signal a bug, and the harness checks for exactly that, making the central claim falsifiable. Finally, serving one JSON contract from both a heavyweight local PyTorch stack and a lightweight hosted vision-LLM engine shows that the *interface* MIRROR tests (predict $\rightarrow$ show evidence $\rightarrow$ explain) is independent of the particular inference backend.

7 Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

Grounded by design. The language layer never sees pixels, only structured evidence, so it cannot invent findings; below-threshold predictions are reported as pertinent negatives rather than dropped, and predictions are surfaced as probabilities to keep uncertainty explicit.



Figure 3: One real study end to end. (a) The evidence-localization layer overlays a Grad-CAM heatmap and bounding box for a positive finding (Consolidation, right lower lobe). (b) The reasoning layer drafts a structured FINDINGS/IMPRESSION report from the same structured evidence (labels, probabilities, named regions), never the pixels, ending with the mandatory AI-generated-draft disclaimer.

Table 2: Predictive quality (DenseNet-121, ChestX-ray14 test split; $N_{\text{test}}=25,596$): per-label and macro AUROC with bootstrap 95% CIs, and macro F1. *Red = illustrative placeholder, pending a trained checkpoint.*

Pathology	AUROC (95% CI)	Pathology	AUROC (95% CI)
Atelectasis	0.771 (0.755–0.787)	Pneumothorax	0.858 (0.843–0.872)
Cardiomegaly	0.884 (0.866–0.902)	Consolidation	0.750 (0.734–0.766)
Effusion	0.831 (0.819–0.842)	Edema	0.849 (0.830–0.868)
Infiltration	0.703 (0.689–0.717)	Emphysema	0.890 (0.870–0.910)
Mass	0.826 (0.809–0.844)	Fibrosis	0.802 (0.775–0.829)
Nodule	0.749 (0.729–0.769)	Pleural Thickening	0.779 (0.757–0.801)
Pneumonia	0.731 (0.703–0.759)	Hernia	0.914 (0.872–0.954)
Macro AUROC 0.806 (0.797–0.815)		Macro F1 0.419 (0.403–0.434)	
Across seeds 0/1/2: macro AUROC 0.805 ± 0.003, macro F1 0.417 ± 0.005			

Table 3: Explanation quality (DenseNet-121, Grad-CAM) against the NIH lesion boxes, over the 8 boxed pathologies. *Red = illustrative placeholder.*

Pathology	#box	Point.	IoU	Loc. acc.
Atelectasis	180	0.572	0.189	0.606
Cardiomegaly	146	0.849	0.342	0.877
Effusion	153	0.660	0.239	0.712
Infiltration	123	0.529	0.177	0.569
Mass	85	0.612	0.224	0.659
Nodule	79	0.430	0.132	0.456
Pneumonia	120	0.650	0.231	0.700
Pneumothorax	98	0.643	0.236	0.684
Overall	984	0.631	0.222	0.675

Score-CAM figures come from the same harness (`--method scorecam`);
[PENDING: run once trained].

Data governance. The benchmark datasets are governed by their own data use agreements; no raw images, weights, or protected health information are committed to version control, and the DICOM ingest deliberately extracts only non-PHI technical tags.

Limitations. The reported numbers are format-illustrative placeholders, not measured results; real training and evaluation on ChestX-ray14 are required before any quantitative claim. Evaluation is currently confined to a single dataset and to chest radiographs; CT and brain-MRI paths are scaffolded but unvalidated. Localization ground truth exists for only 8 of 14 pathologies. Report quality is presently assessed qualitatively; a quantitative comparison against real radiologist

reports (for example MIMIC-CXR [4]) is future work. Class-activation maps are coarse and can highlight context rather than the lesion itself [14].

8 Conclusion

MIRROR composes classification, evidence localization, and grounded report generation into a single traceable chest-radiograph pipeline, paired with an evaluation framework that measures prediction and explanation quality side by side and, via an ablation that holds predictions invariant, frames the contribution as interpretability added at no predictive cost. *[PENDING: state the measured outcome and the ablation result once the model is trained.]* The immediate next step is to replace the illustrative results with real numbers from a trained checkpoint, then extend the same evidence-grounded recipe to the CT and brain-MRI modalities already scaffolded in the codebase. The system and its evaluation harnesses are released open-source to support reproducible study of interpretability and trust in medical imaging.

Reproducibility

Code: <https://github.com/vignesh-nagarajan-vn/MIRROR>. Live demo: <https://mirror-ten-jet.vercel.app/>. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the full pipeline on ImageNet weights with the offline template backend, with no checkpoint or API key required. All evaluation harnesses live in `evaluation/`; every results JSON

Table 4: Ablation (DenseNet-121). Predictive metrics are *identical* across rows (max probability delta = 0; the added layers are post-hoc), so interpretability is added at no predictive cost; localization capability and its latency appear only where enabled. *Red = illustrative placeholder*.

Condition	Macro AUROC	Macro F1	Pointing game	Mean IoU	Latency (ms)
Classification only	0.806	0.419	n/a	n/a	40.9
+ Localization	0.806	0.419	0.631	0.222	173.2
Full MIRROR (+ report)	0.806	0.419	0.631	0.222	174.4

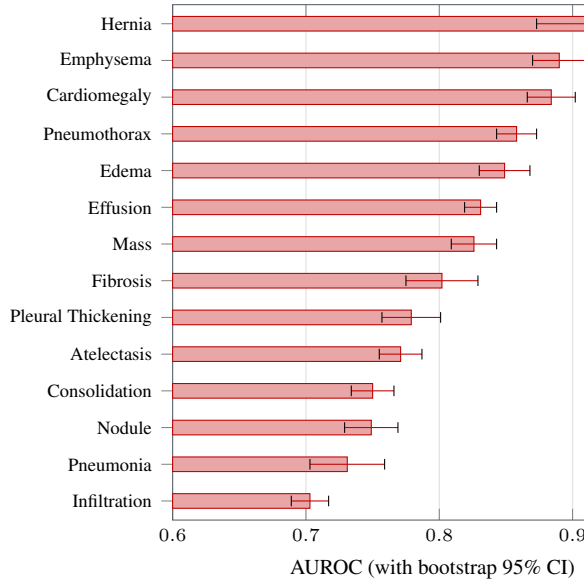


Figure 4: Per-label predictive quality: AUROC with bootstrap 95% CI whiskers, sorted ascending. *Red = illustrative placeholder data, pending a trained checkpoint.*

stamps a `reproducibility` block (seed, git commit, library versions), and the unit tests are torch-free so they run without a model or dataset.

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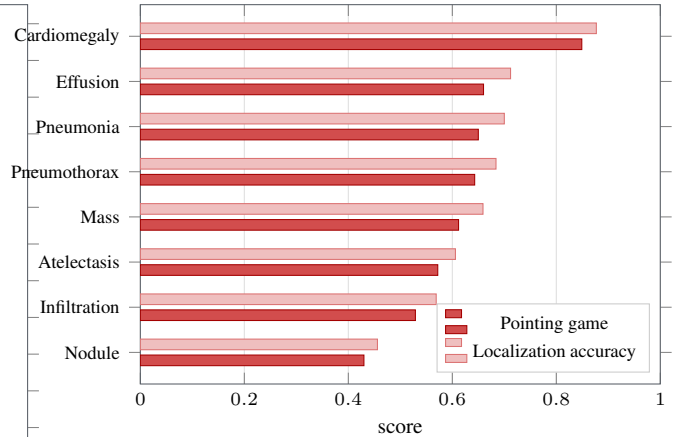


Figure 5: Per-lesion explanation quality: pointing-game accuracy and localization accuracy ($\text{IoU} \geq 0.1$) over the 8 boxed pathologies, sorted ascending. *Red = illustrative placeholder data.*

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